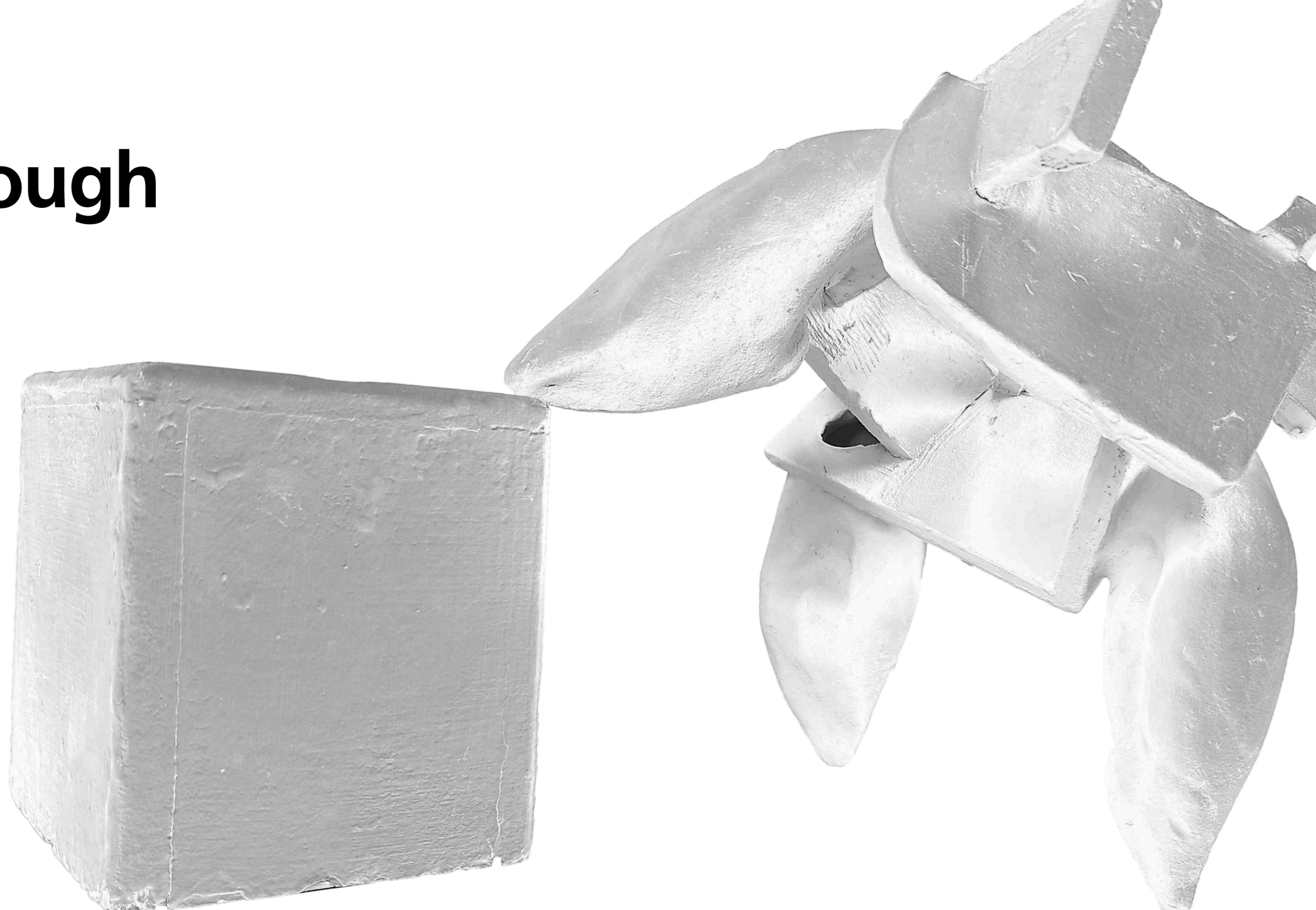
Brom McCullough

Industrial design portfolio



Outdoor Bench

Commissioned furniture

June 2026

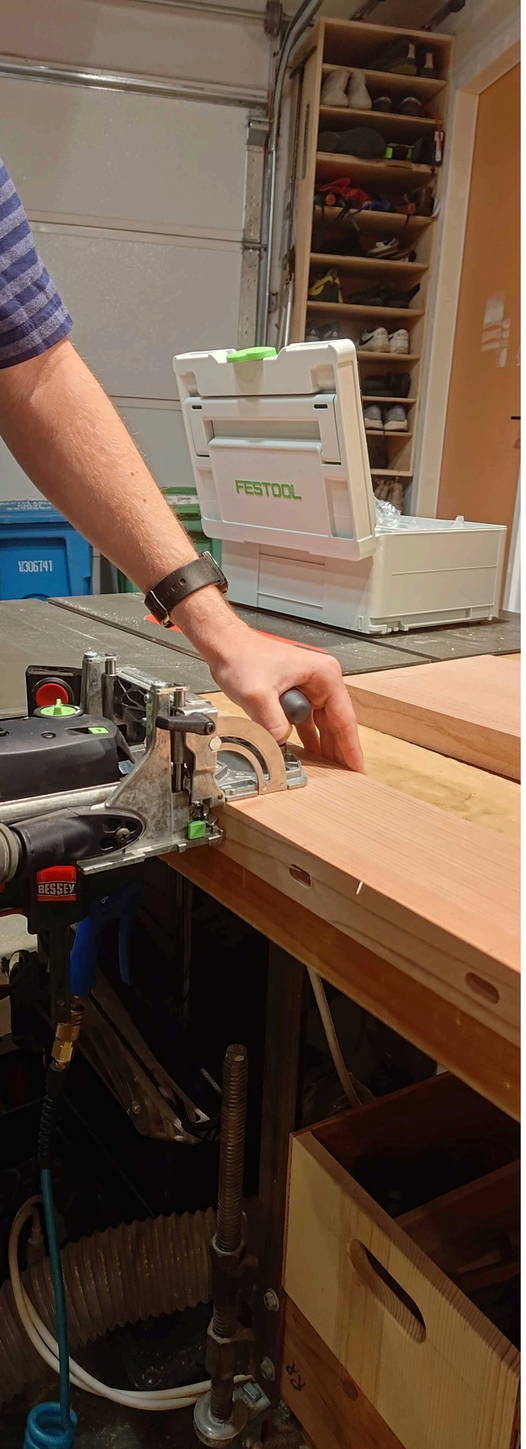
Initial
doccumentaion shot

Image in place at
clients house

Base glue up.

After application of Waterlox exterior oil finish.

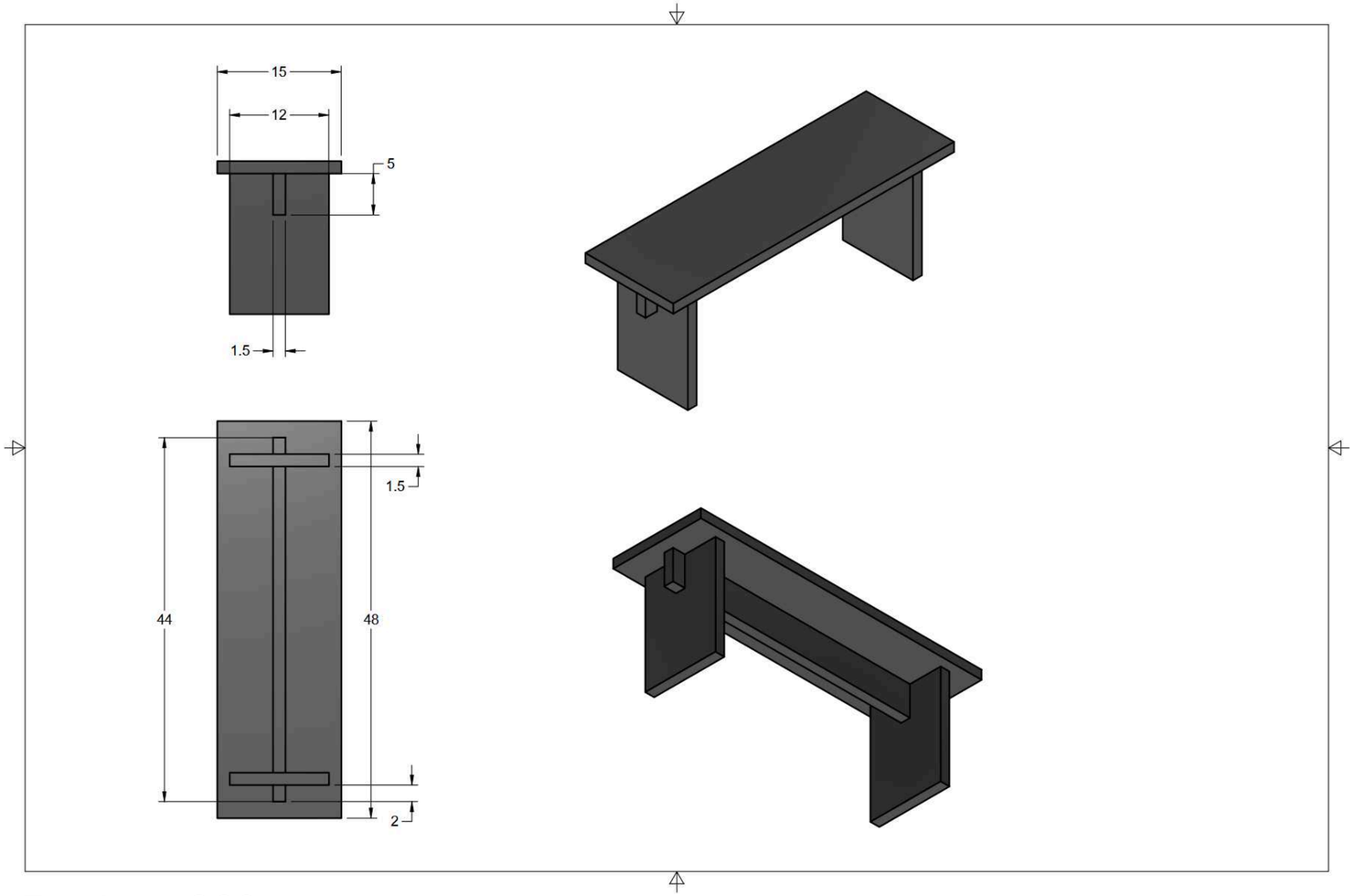
Installing of plastic feet to keep end grain off ground.



Domino joinery of a bench seat plank (left), glue up of 3 bench seat planks (right).

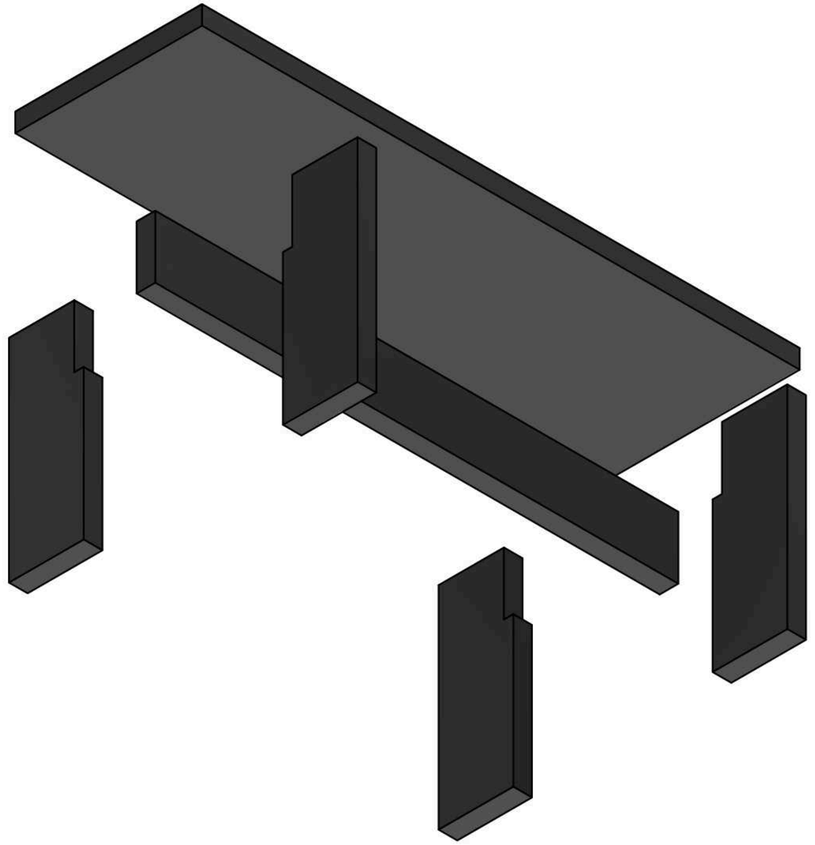
Planks for bench seat, legs and support beam.

Climb and typical router cut around perimeter of internal face to prevent grain blowout.



Fusion 360
layout
Units: Inches

Bench needed to take shoes off at clients gate. Proportions finalized around body proportions of household members.



Exploded view

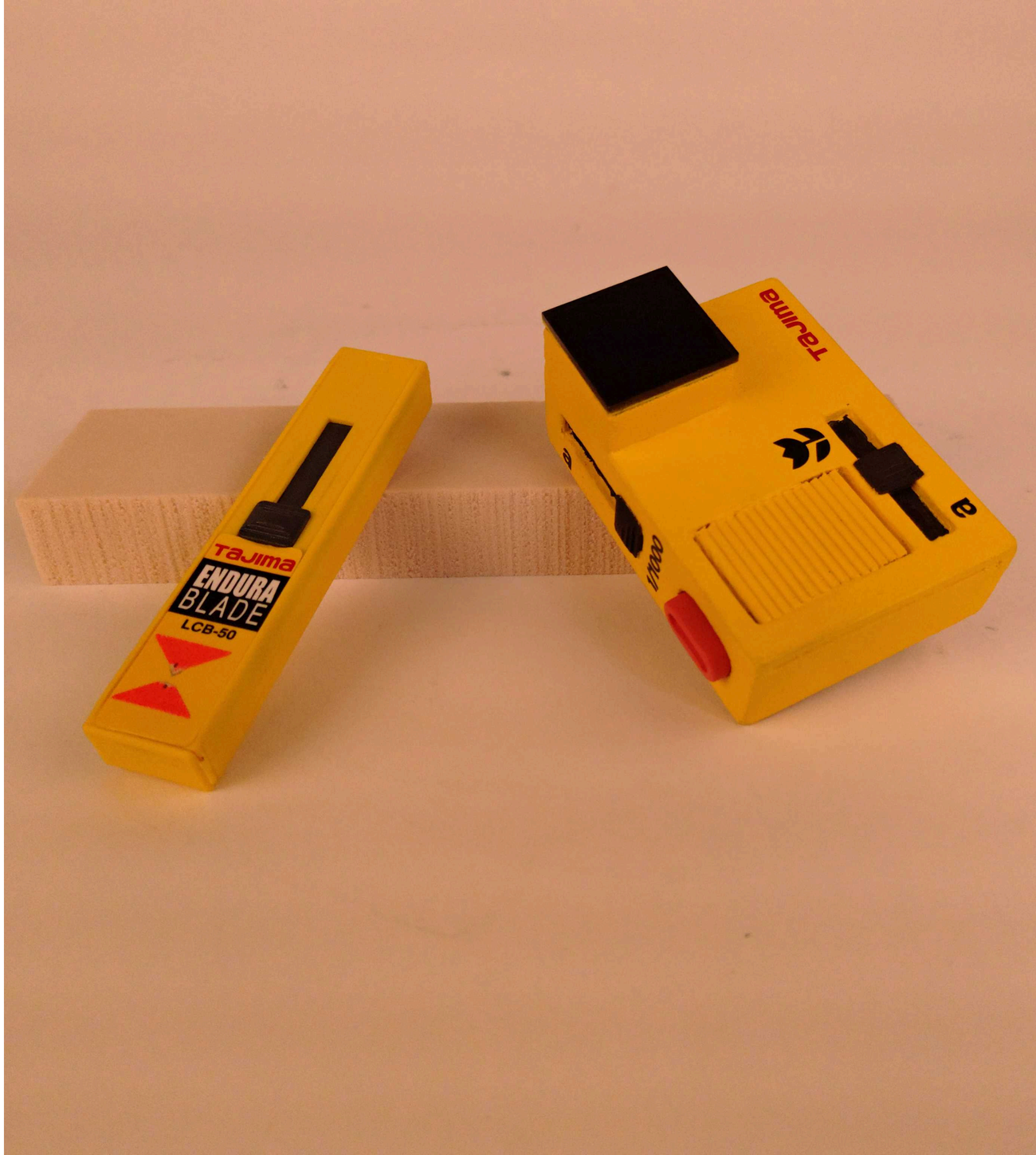
Base components laid out before glue up, domino and dowel joinery used.

Domino joinery of bench seat and base.

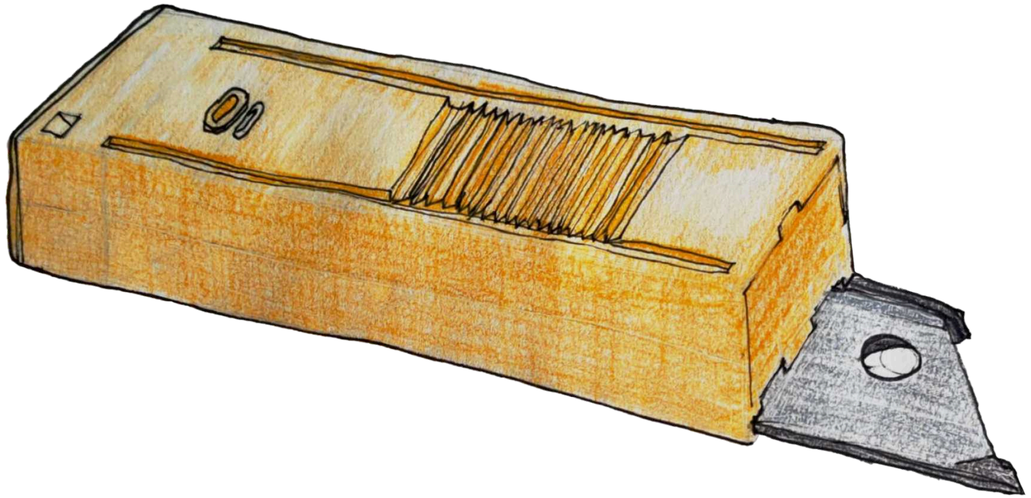
Tajima Endura Photo

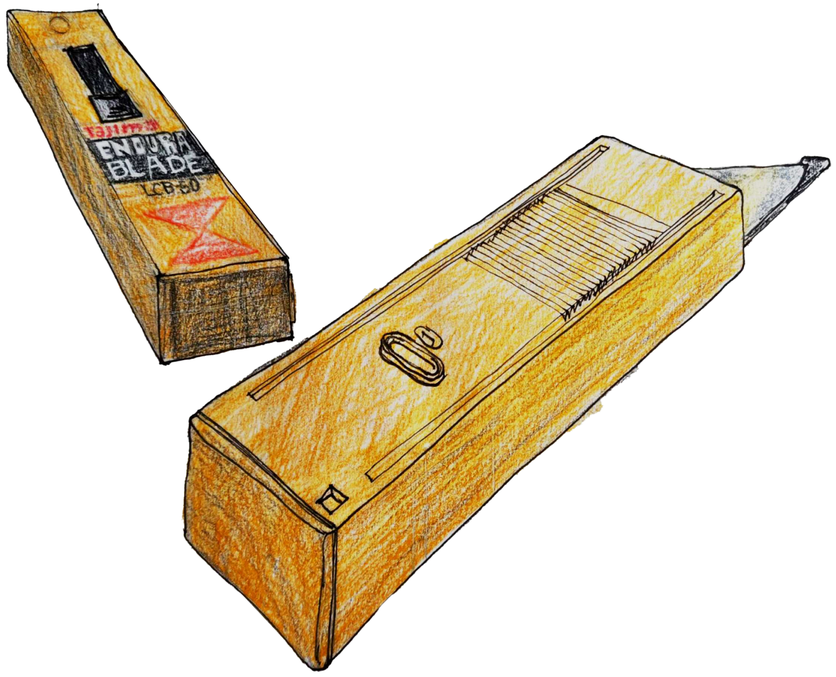
Product line extension

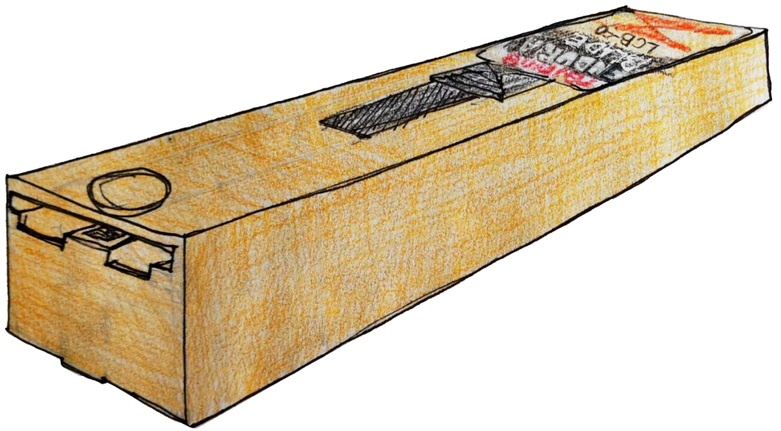
March 24th – April 30th
2025

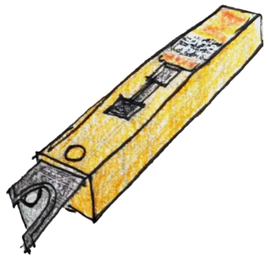


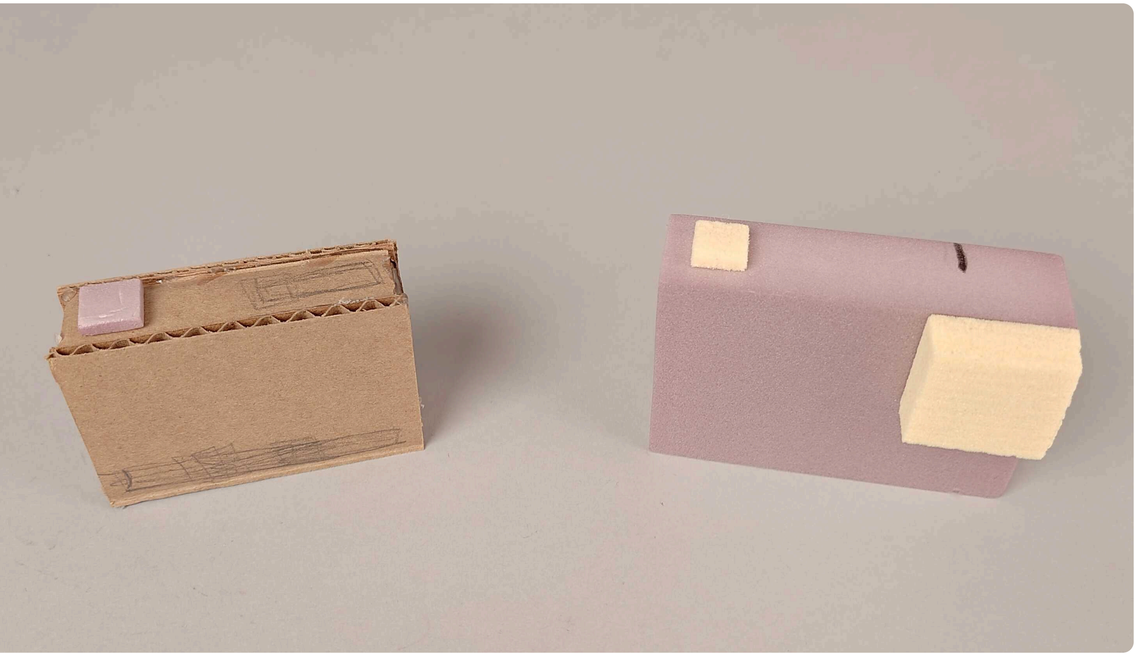
Project to extend the Tajima Endura blade brand into a new product. The simple injection mold details applied well to the camera form. Making placing features for a new product easy.

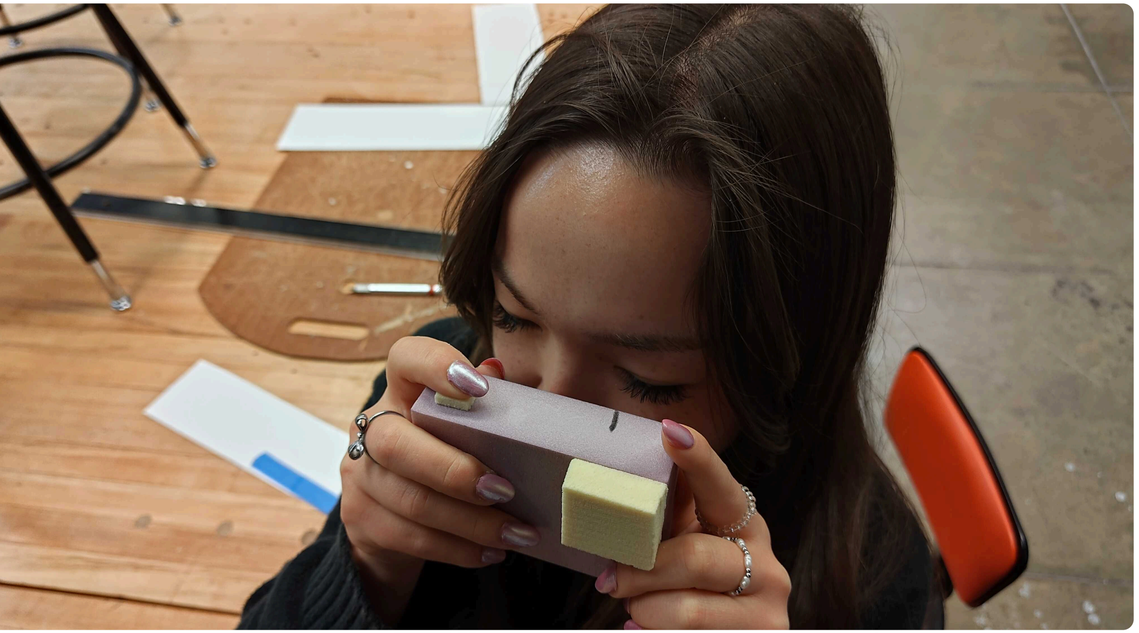






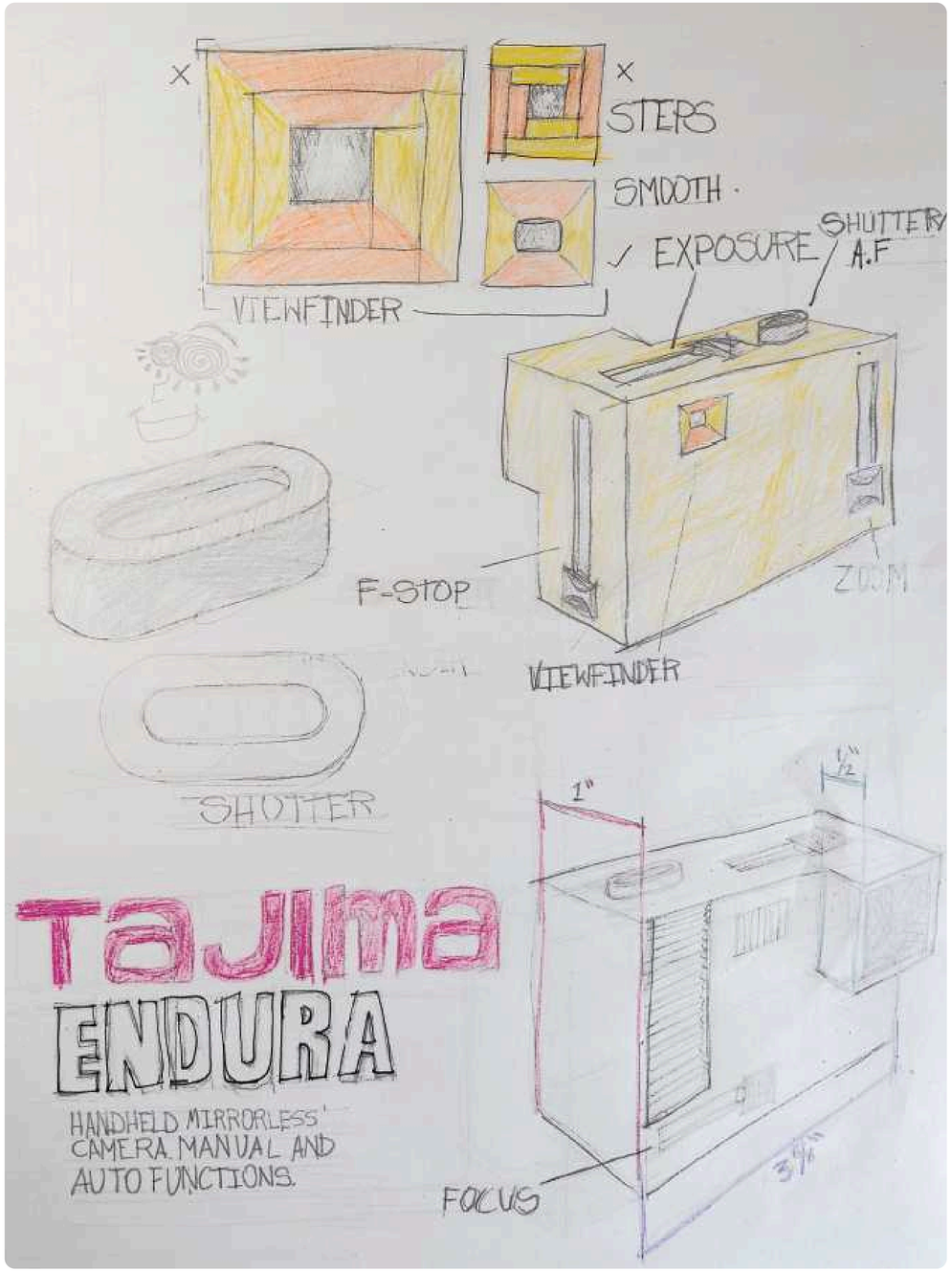


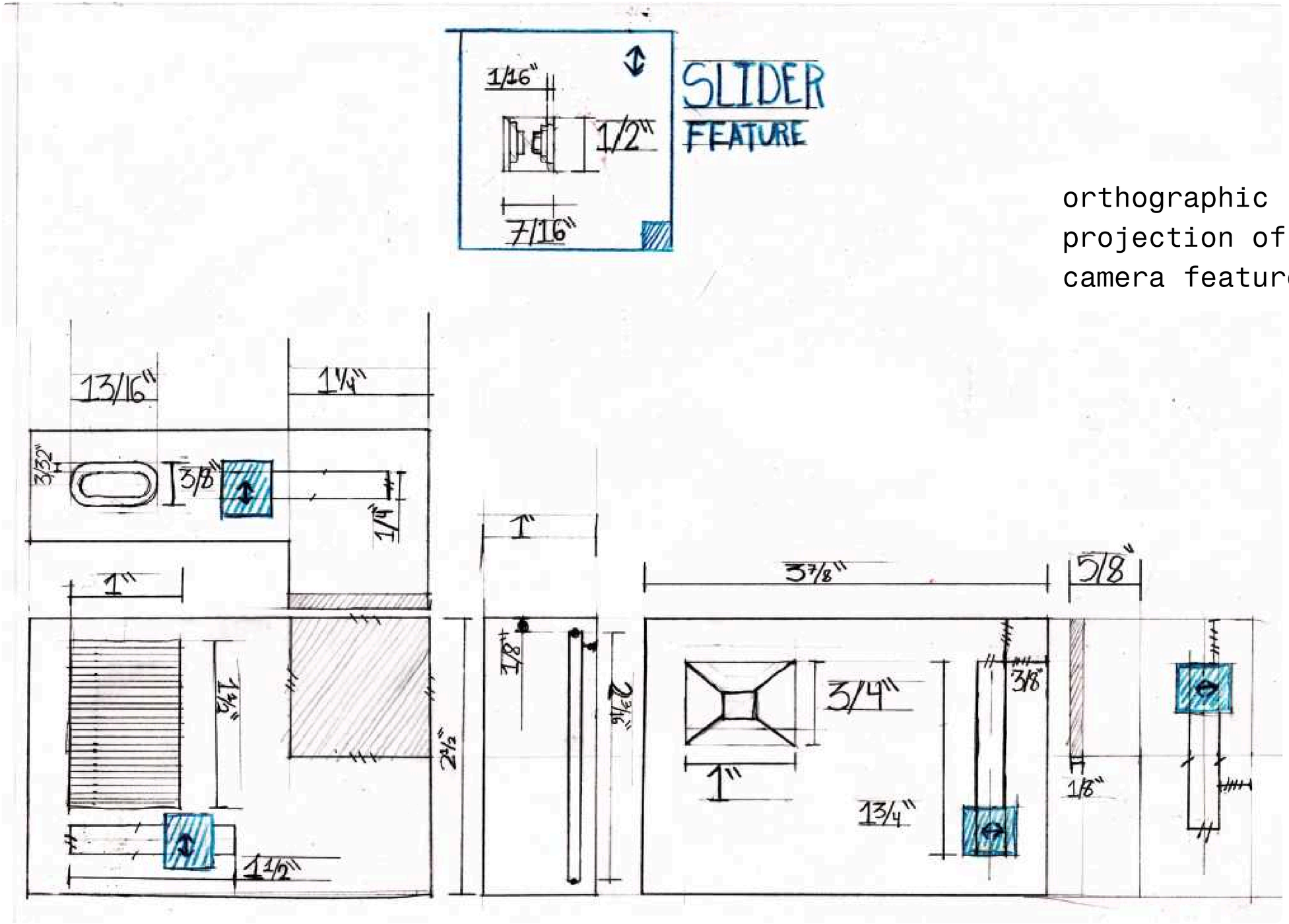




User testing feature placement and size limits solved many questions making aesthetic implementation easier. My sample group included individuals at either extremes of hand size and familiarity with cameras. Individuals who had familiarity with digital cameras and small or large hands provided especially useful feedback. This finalized placement of all features. A/F & shutter button various setting sliders, and the camera protrusion.

After the 2 low fidelity models were used to test function and form, I sketched out and drafted the dimensions in an orthographic projection to place the shutter button, viewfinder, and sliders for various camera functions.





orthographic projection of camera features

Assembly of
final PU foam
and bondo model.
Imaged to right
inlaid 3d
printed grip.

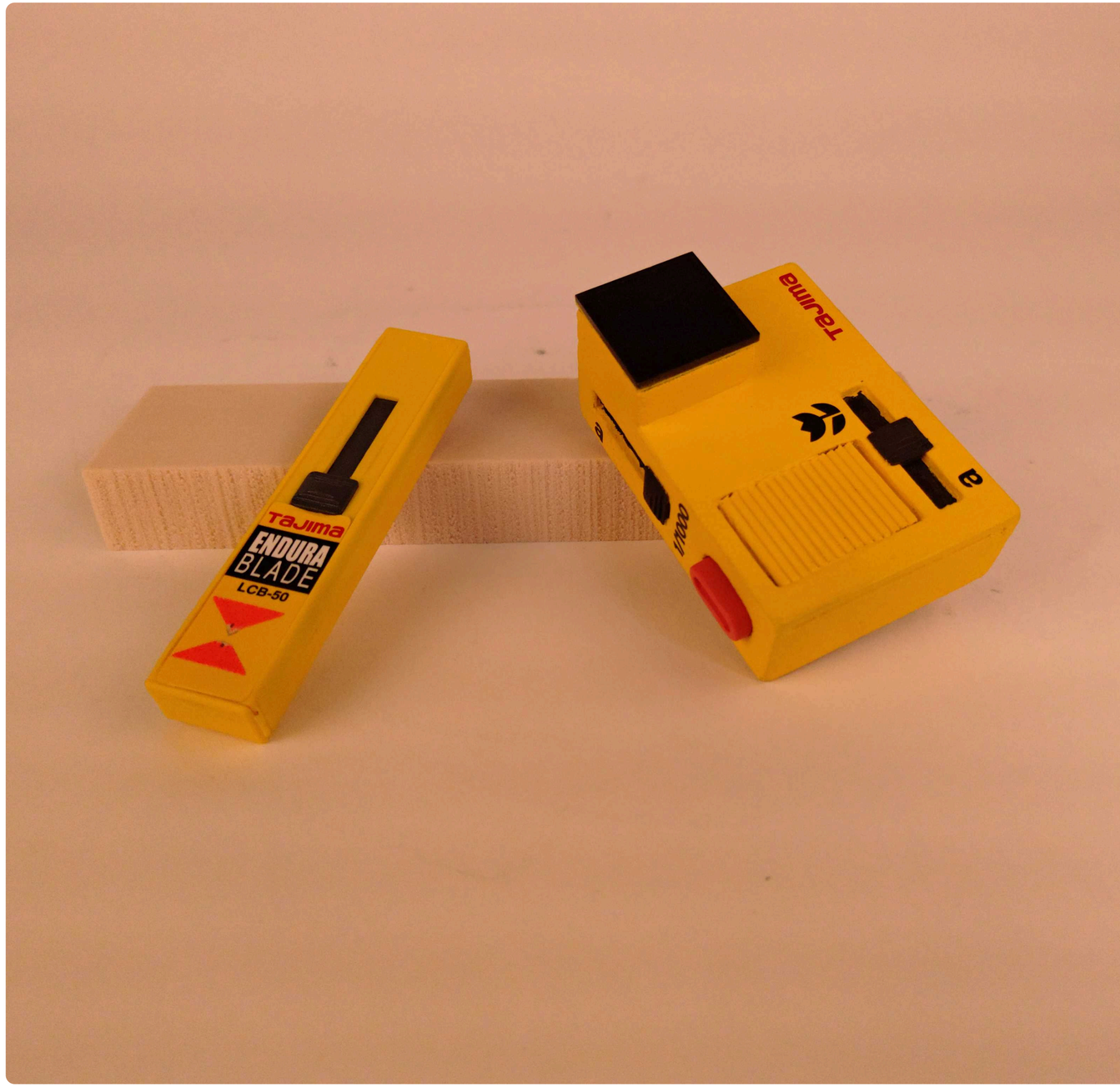
Final model was clean, bright and adhered to the design language of the source product. I learned order of operations to model with PU foam, Bondo, and spray paint.

Comparison between Endura Blade products. Shutter buttons come in two colors, orange and black. Samples of Logos used for labeling slider functions.

A B 1/2

ENDURA PHOTO DSLM-50 ƒ3.3

1/1000 ƒ6.3



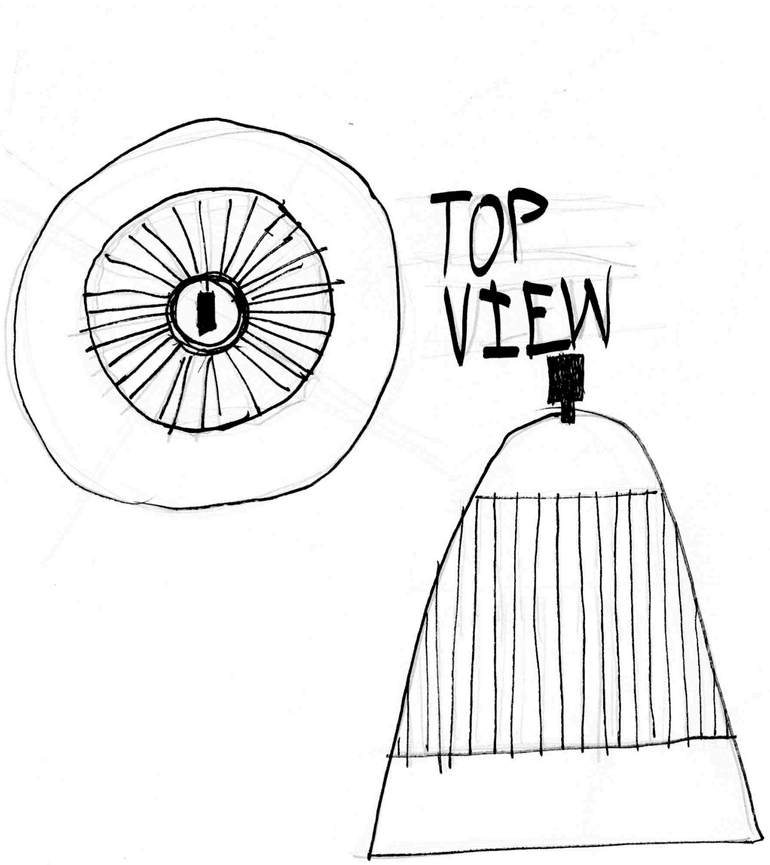
Linnticular FilterLight

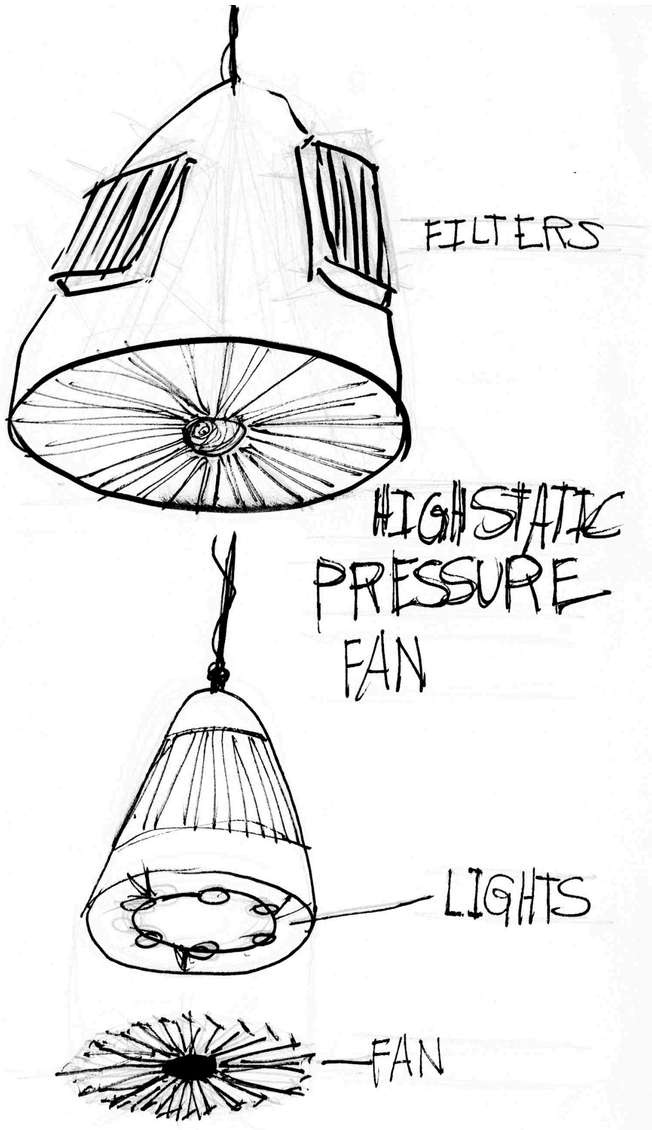
Concept product

March 24th – April 30th 2025

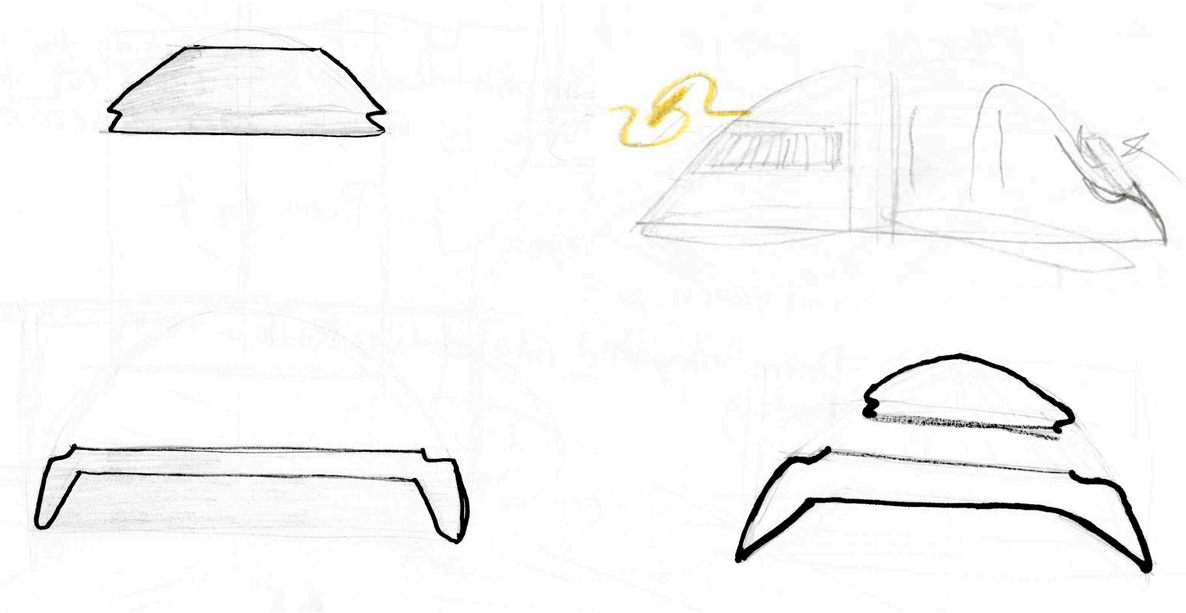
Modeling, Rendering

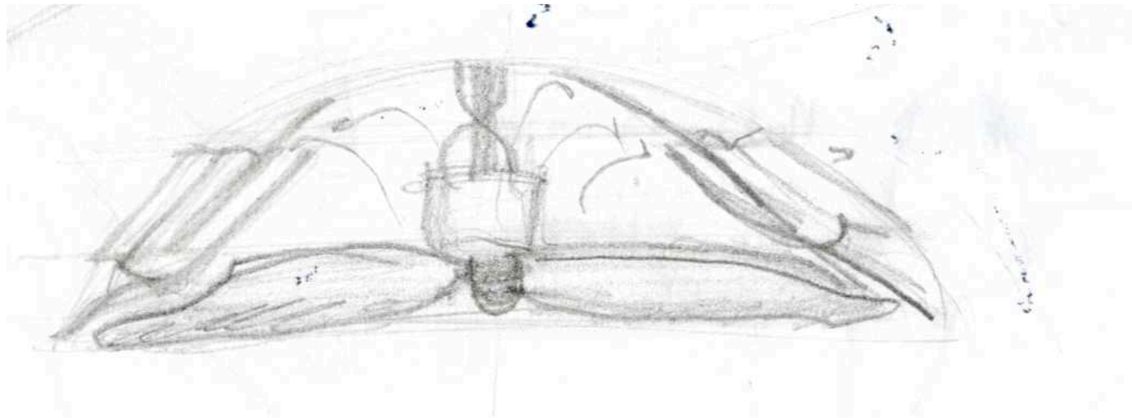
A surface modeled conceptual air filter/room light combo designed to provide ventilation in living rooms and large shared spaces. The filter must be replaceable without releasing dust into the environment when disturbed or replaced.

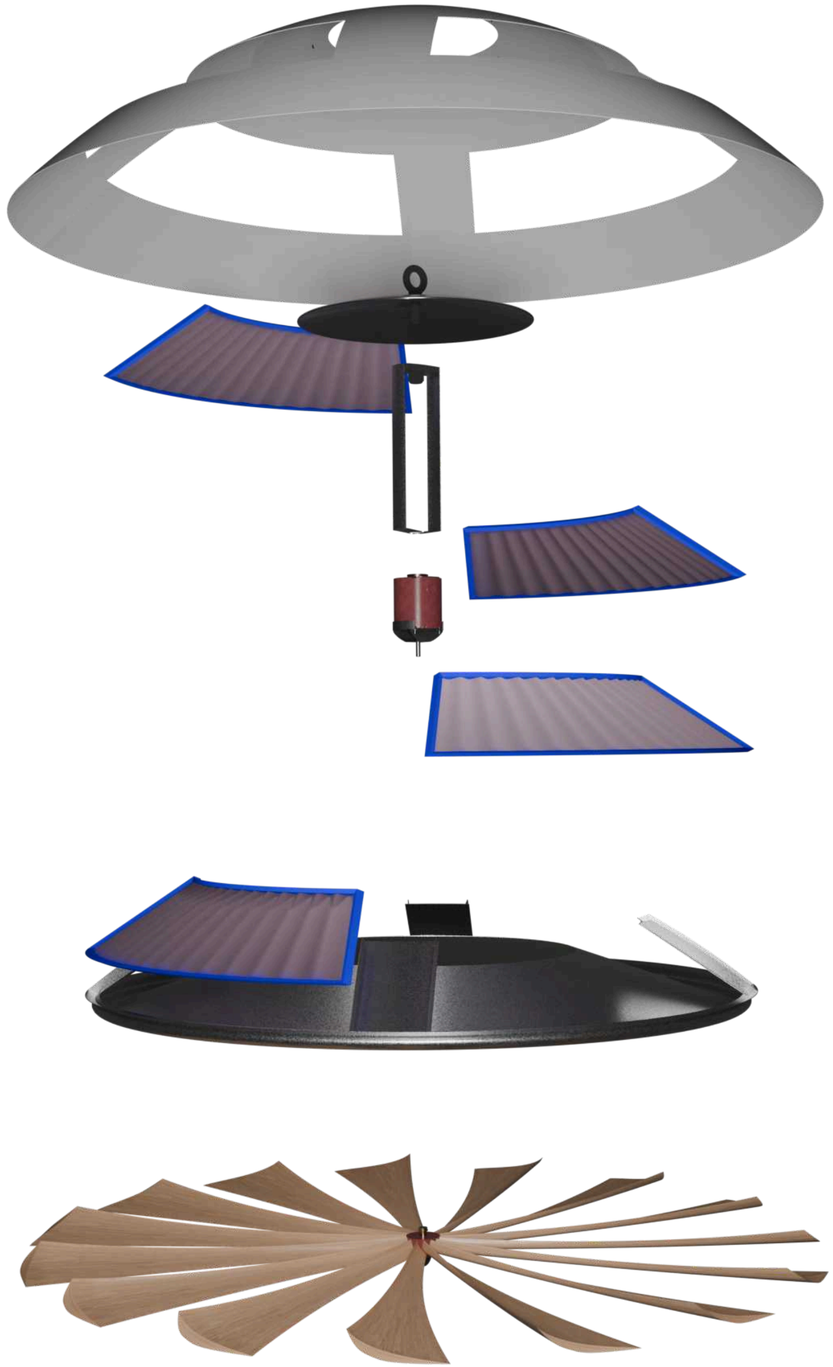




Sketches firming up internal features and ducting. Pushing air up through the internal side of filters provide air filtration. Internal light sources cast through filters diffuse light.

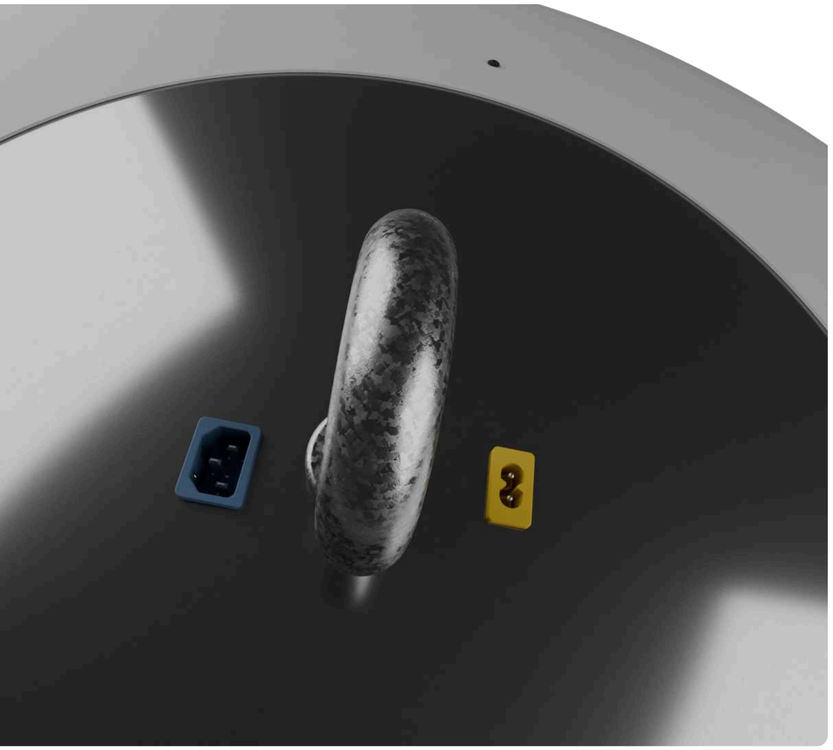


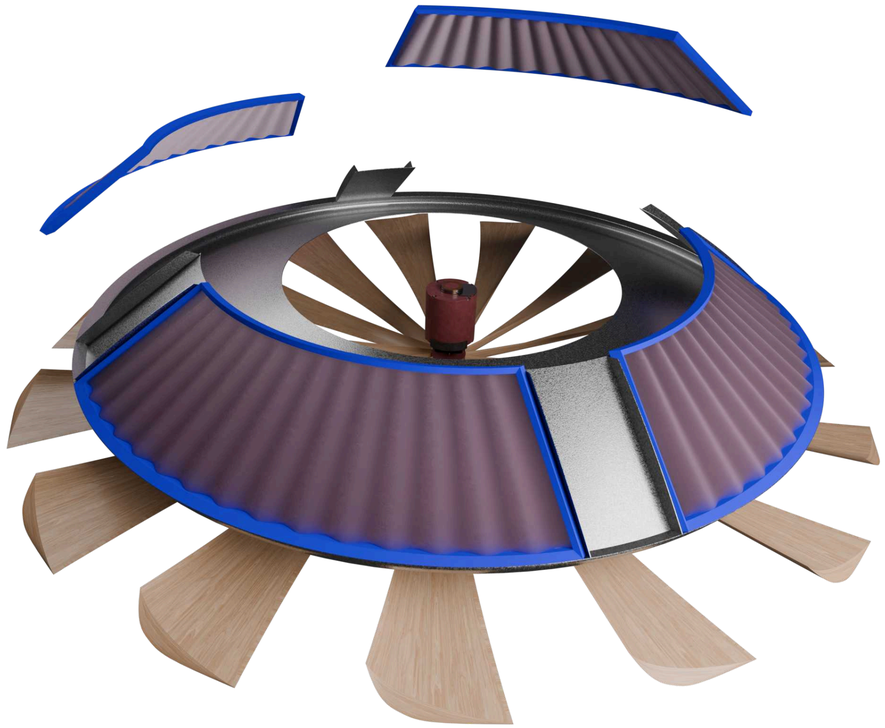


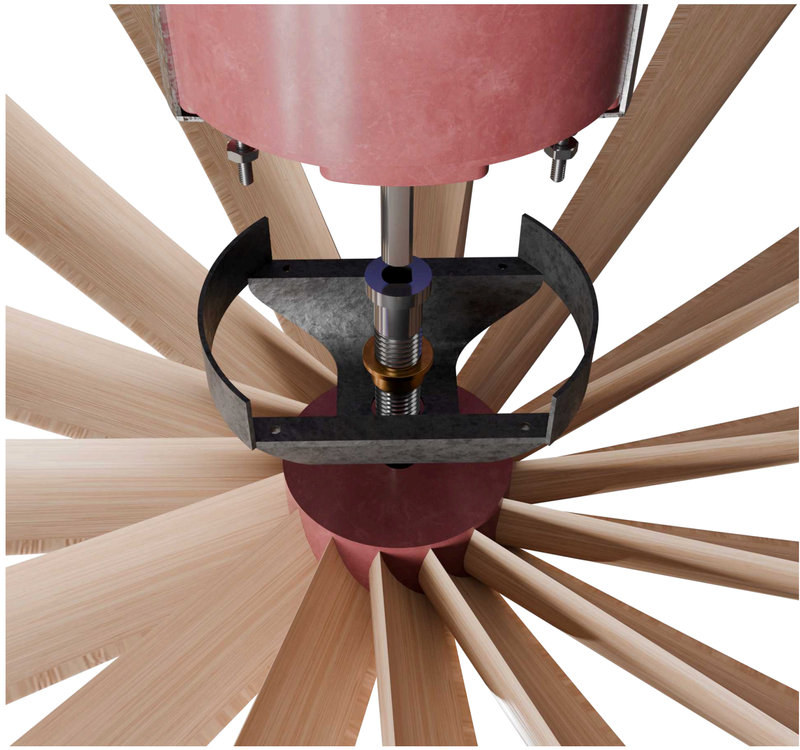


My model is designed for welded, turned and crimped sheet metal. Component geometry and fitment was difficult to surface model in rhino without parametric modeling.

I replicated parametric modelling techniques in rhino. I backed up every step that permanently changed surfaces, geometry, and sketches. This rapidly ballooned the file size and file structure complexity however mistakes were easy to revert and rebuild.







Close up shots of Galvanized ceiling eye bolt and ac connectors, exploded duct filter assembly and fan thrust bearing threaded mount.

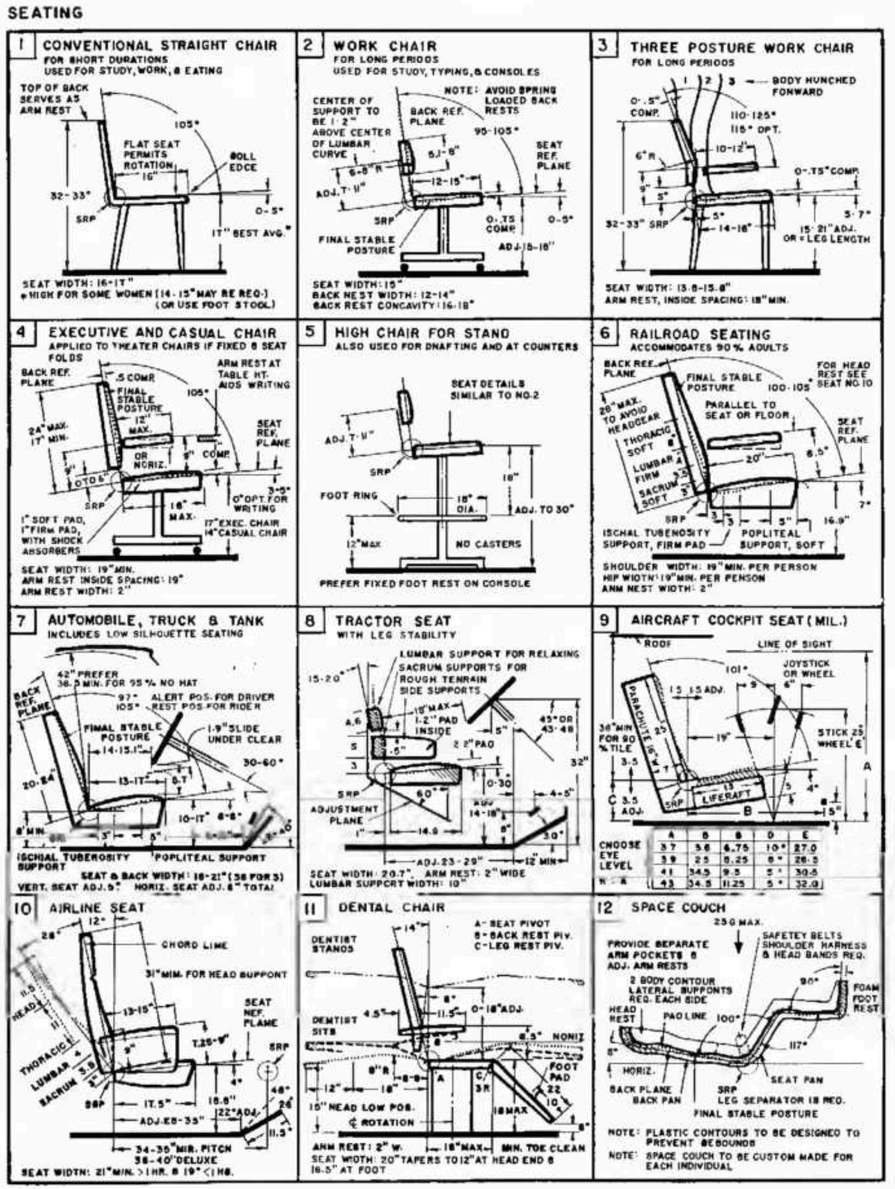
The gestural strengths of surfacing modelling was particularly well suited for certain aspects of this design and extremely difficult in others. Joining crimped sheet metal features had unique drawbacks with surface modelling. A blend of parametric and surface modelling would fit this problem uniquely, however I lacked time to figure out and implement a parametric and surface modeling workflow in this project.

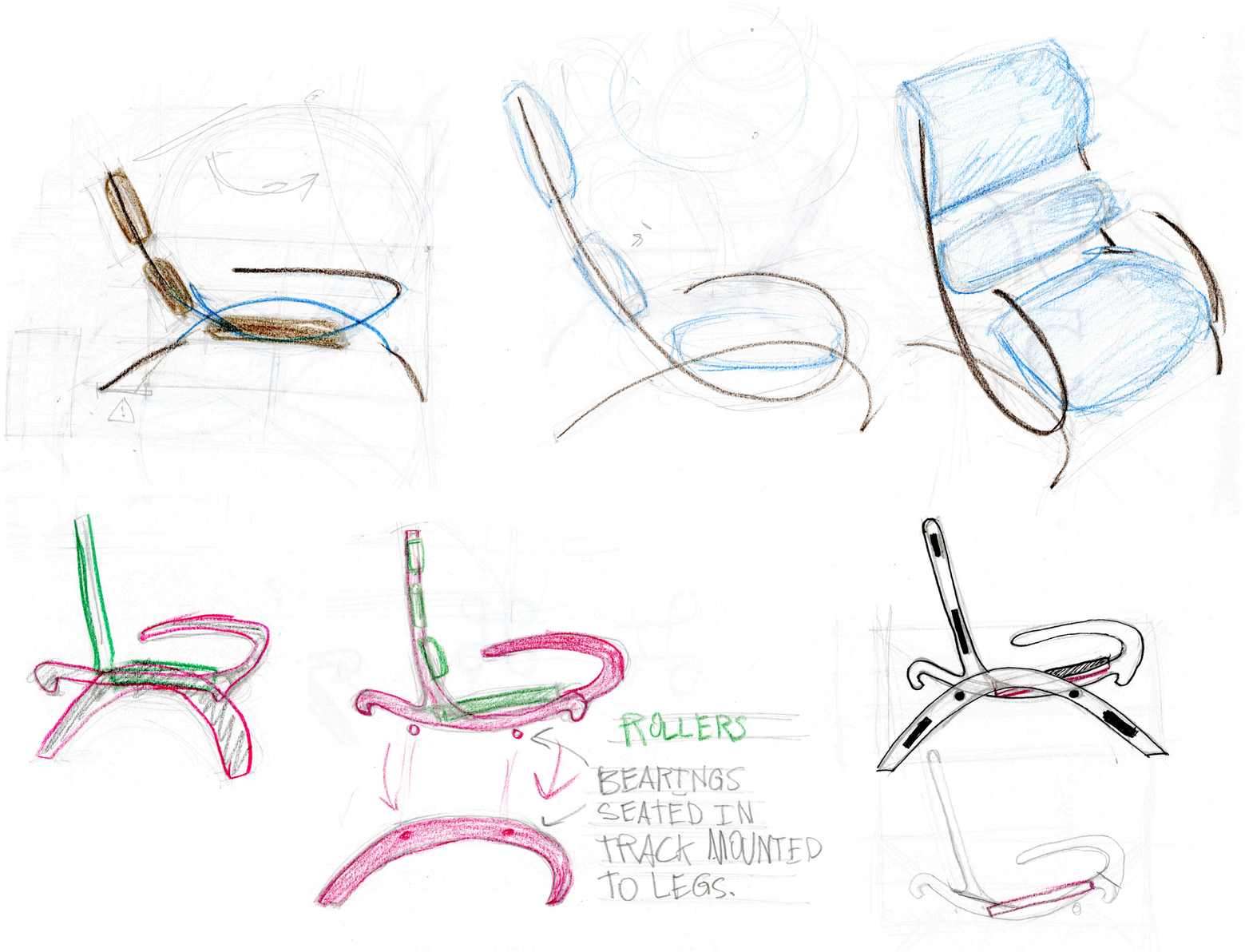
1/3 Recliner

CNC chair project

March 24th – April 30th 2025

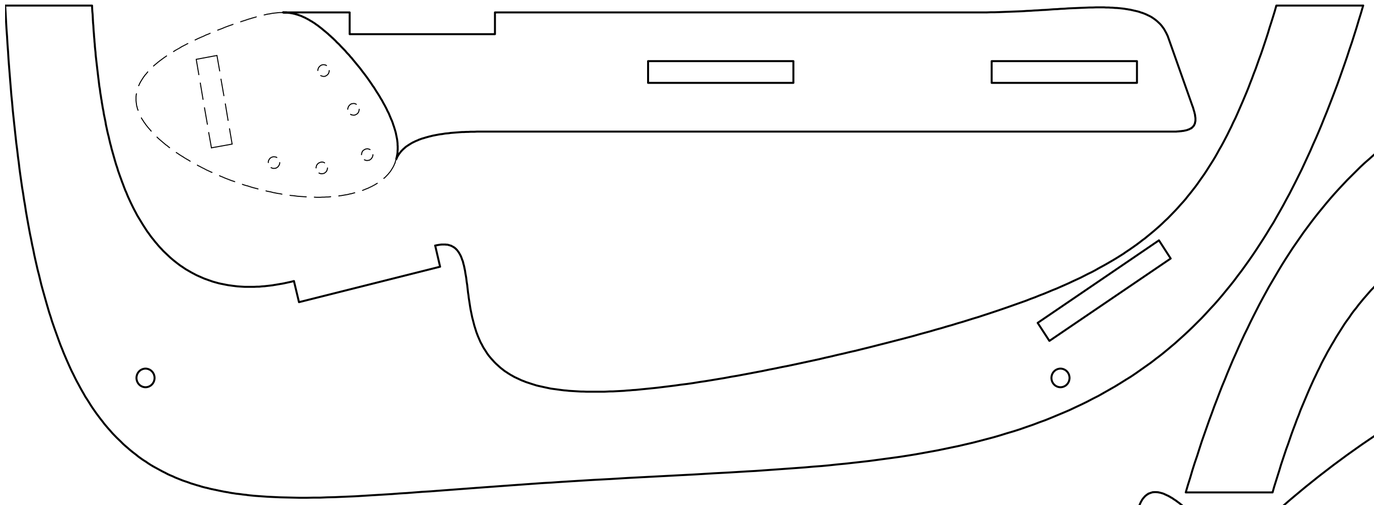
This recliner project had to be made out of a single sheet of birch plywood with a CNC router. I wanted to make a recliner that was relatively comfortable in both a prone and relaxed position while being as simple and easy to construct as possible. I used the airline seat dimensions from the 1966 study Measure of Man ergonomic study by Henry Dreyfuss.

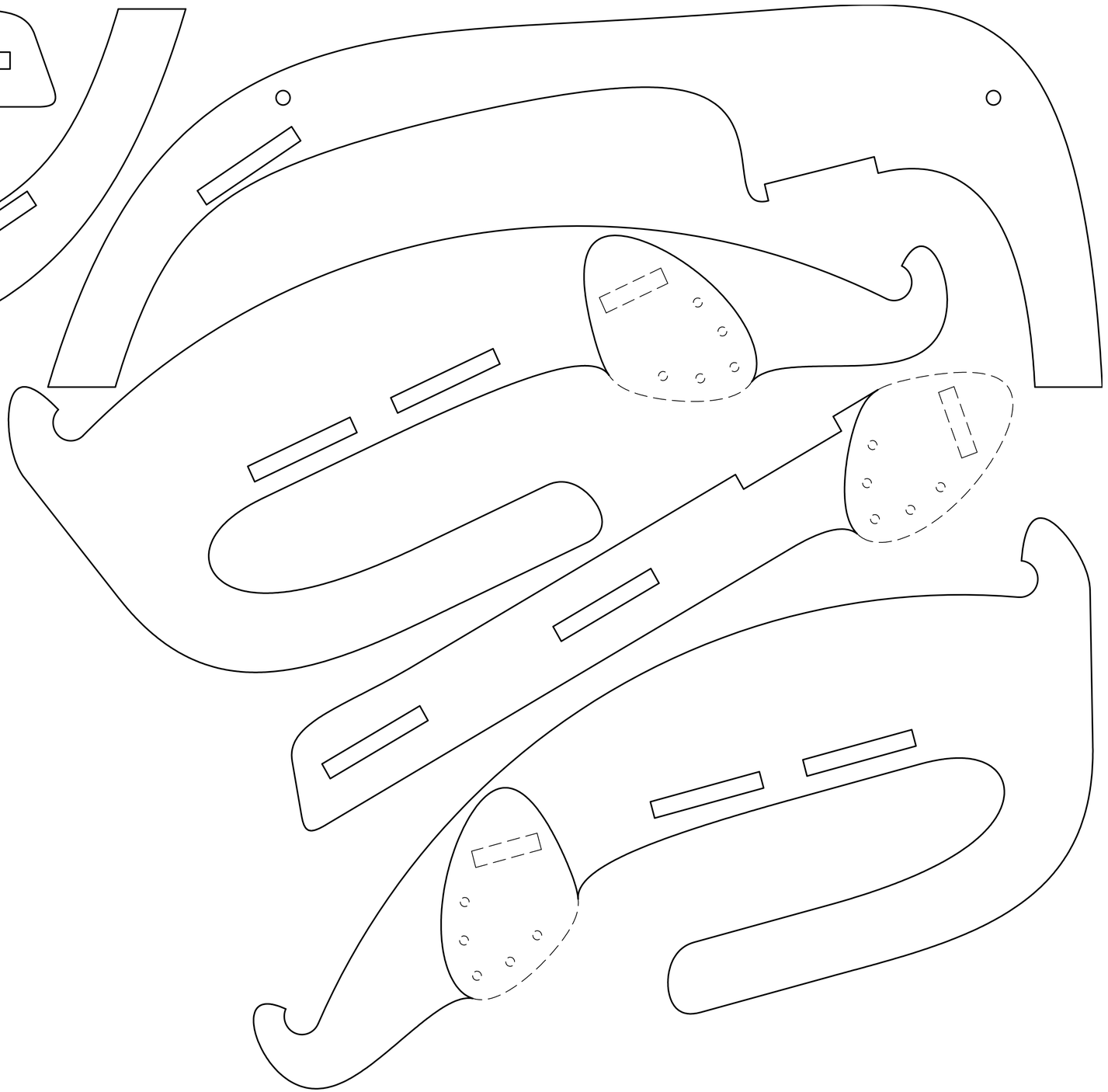




Ideation and semi technical sketches

Scaling and testing the parametric model to hone proportions was rapid allowing quick development.

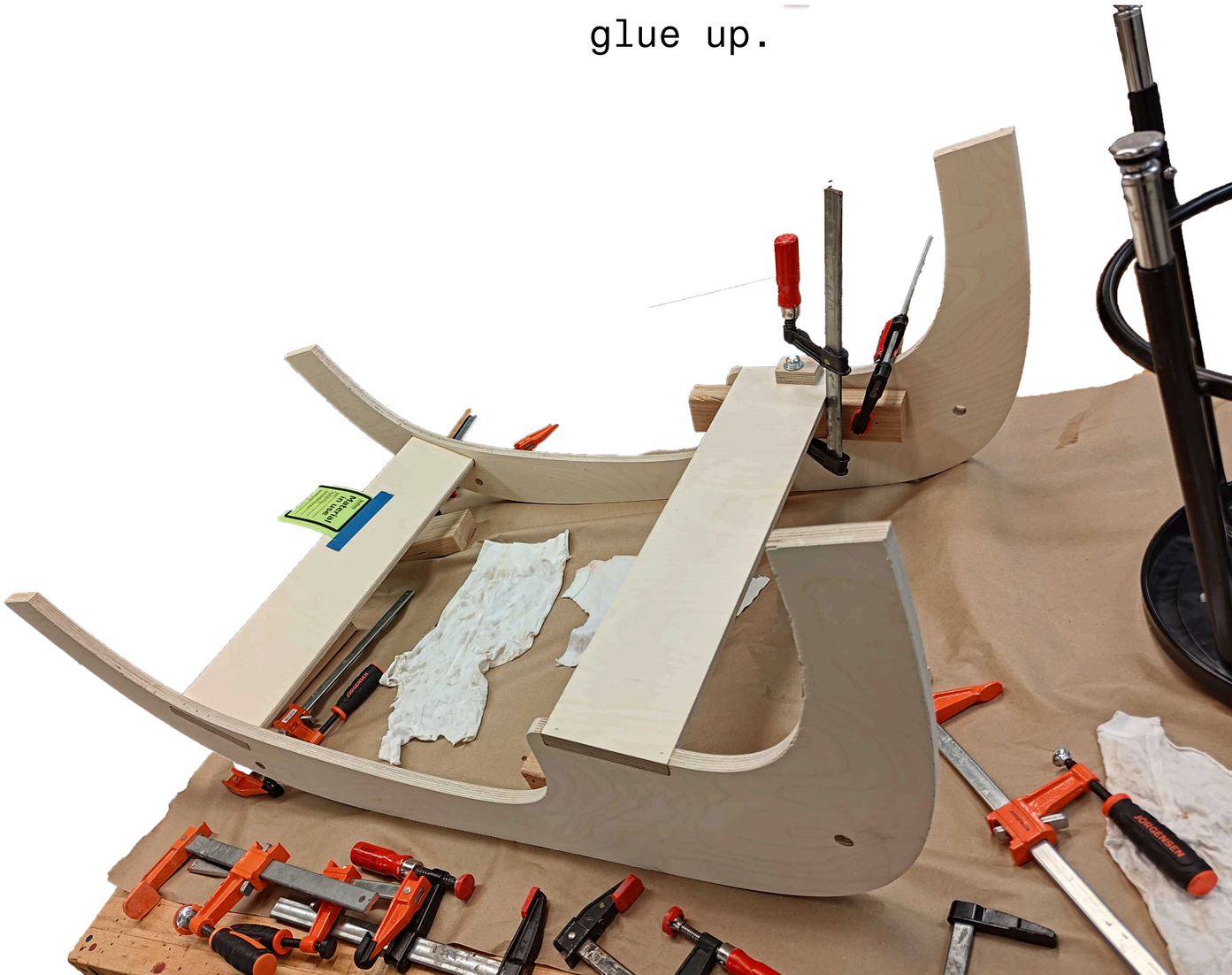




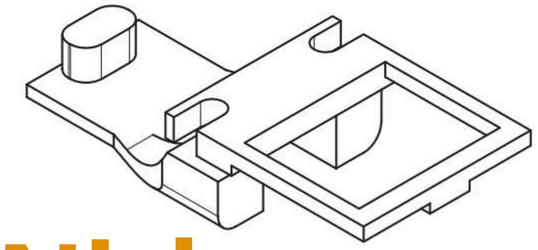
Getting the recliner to fit into the plywood footprint was the first challenge. The travel of the chair had to decrease so it would sit on a single sheet.

There were issues tolerancing slats and general fitments. The intention was easy press fits, making for an easy glue up. The slats were too loose the 2 side halves were too tight. Post processing and elaborate glue ups solved these issues.

Cradle and Seat
glue up.



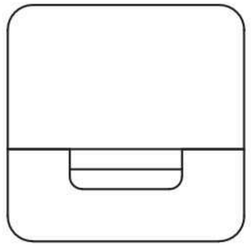
Bearing assembly for range of motion constructed of bolts, thread lock nuts, steel washers and brass spacers.

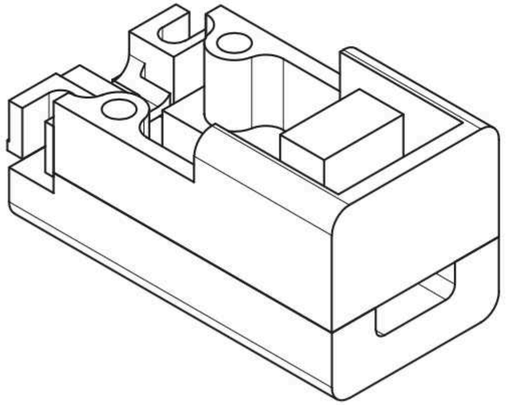


Nichrome Lighter

Circular senior design project

Spring 2026

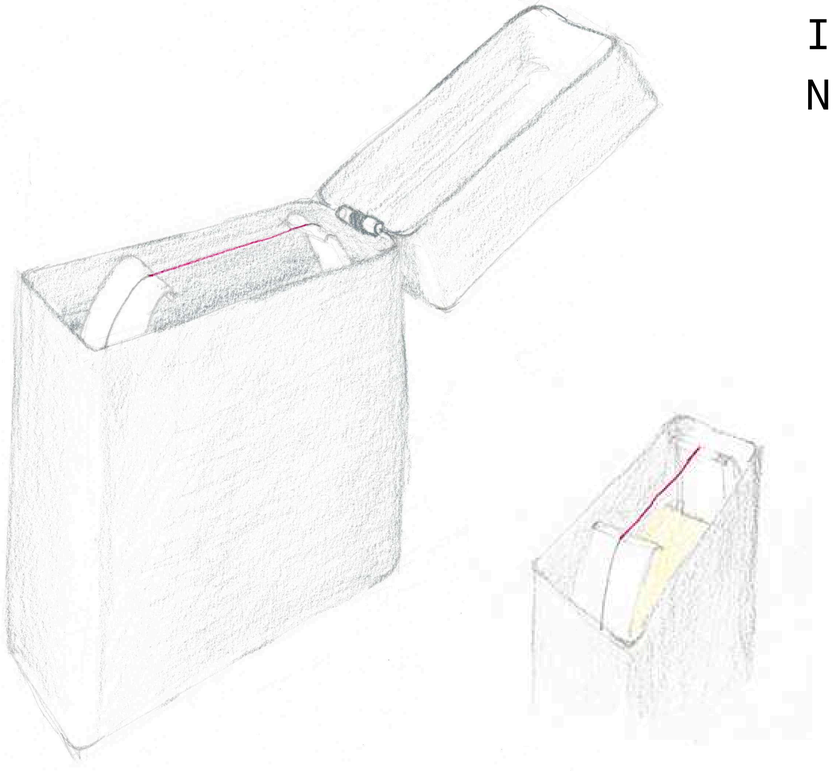




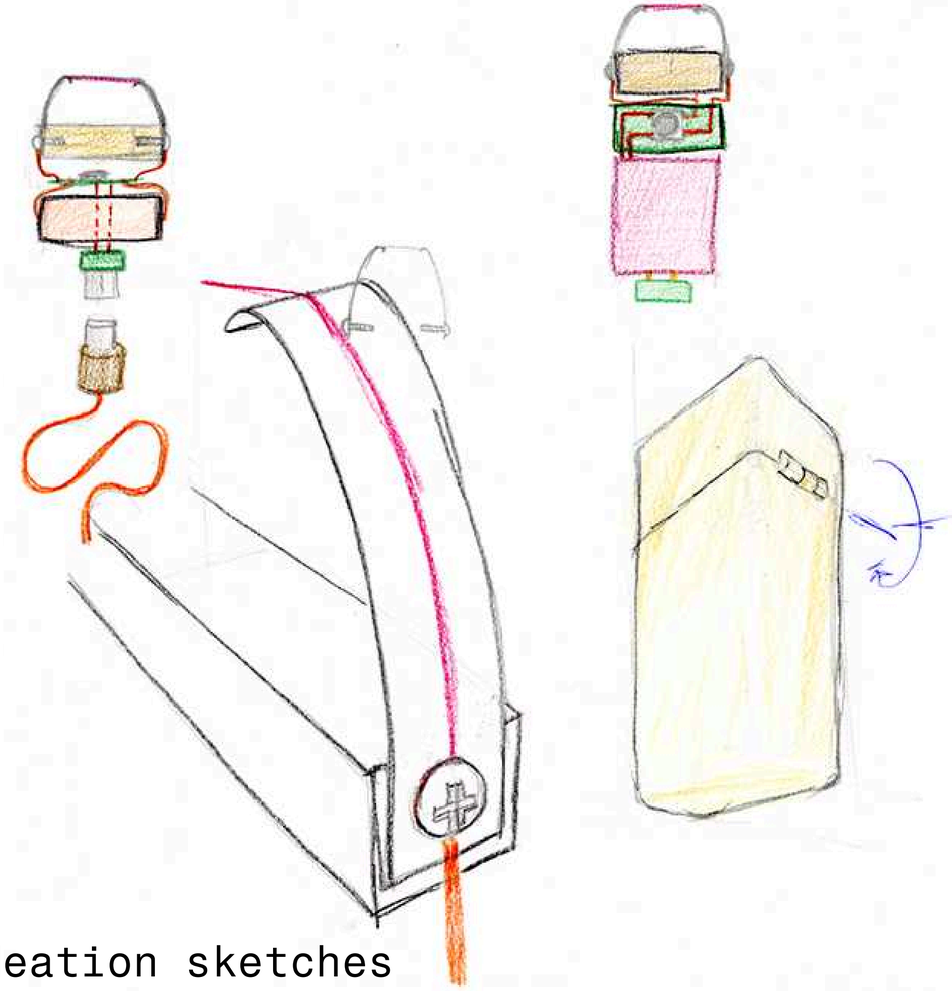
As my capstone project i wanted to combine my interest in user centric design and post consumer materials. I chose to combine both into a product to explore a product that can participate in a circular economy.

After a literature review of economic, waste management and design articles I distilled 3 values for a successful design:

1)discrete and separable components,
2)Use widely available components when possible,
3)Insure product can be disassembled with basic tools.



I chose a lighter that uses Nichrome wire and discarded vape batteries as the exploratory product as it was not too complex and within my technical skillset to complete.

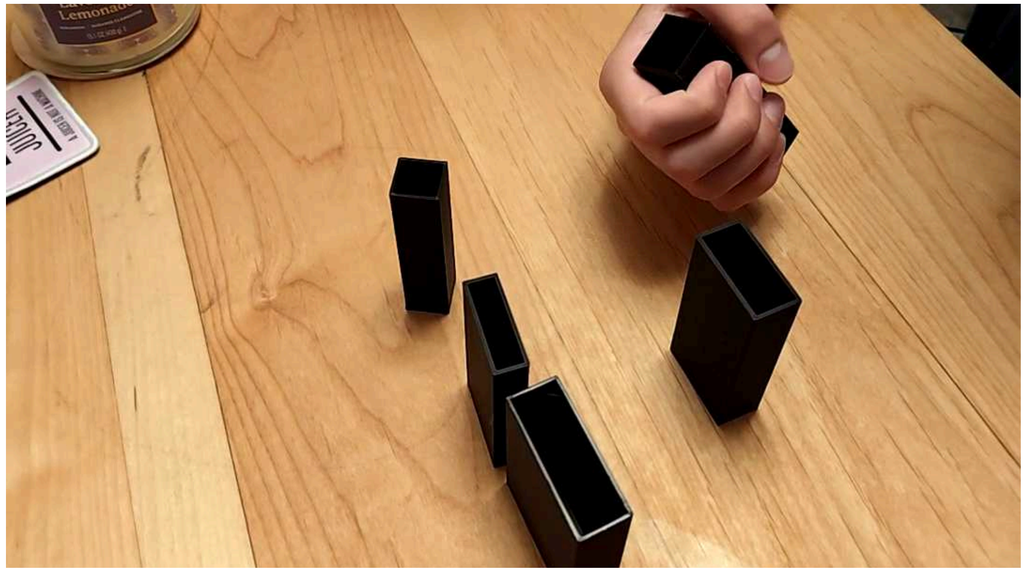


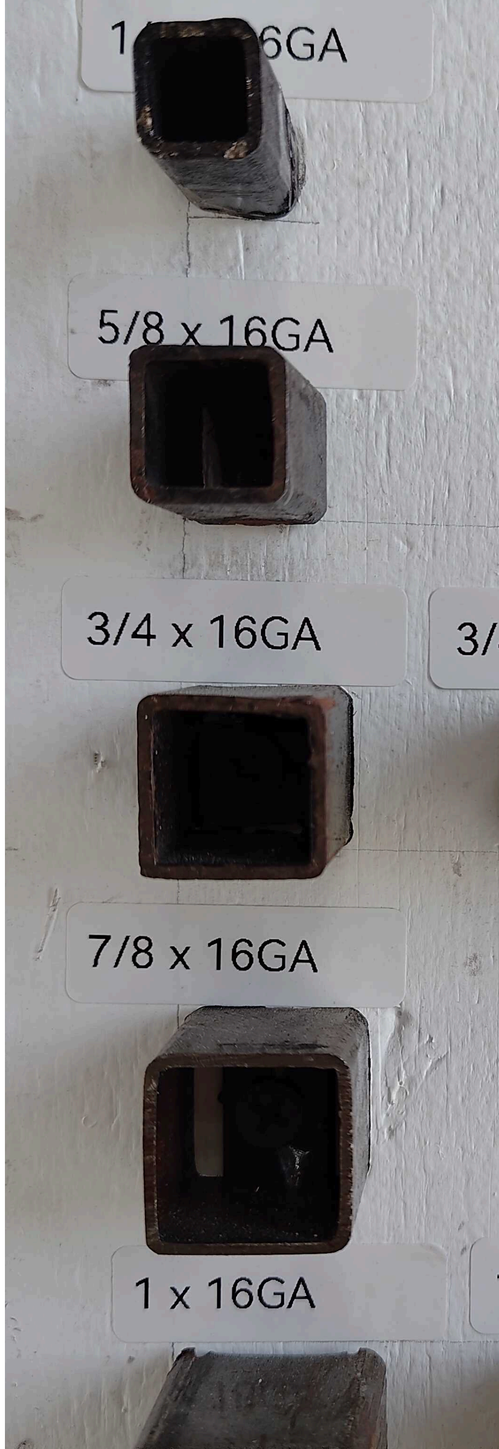
Ideation sketches of components, fitments and enclosures.

User Testing

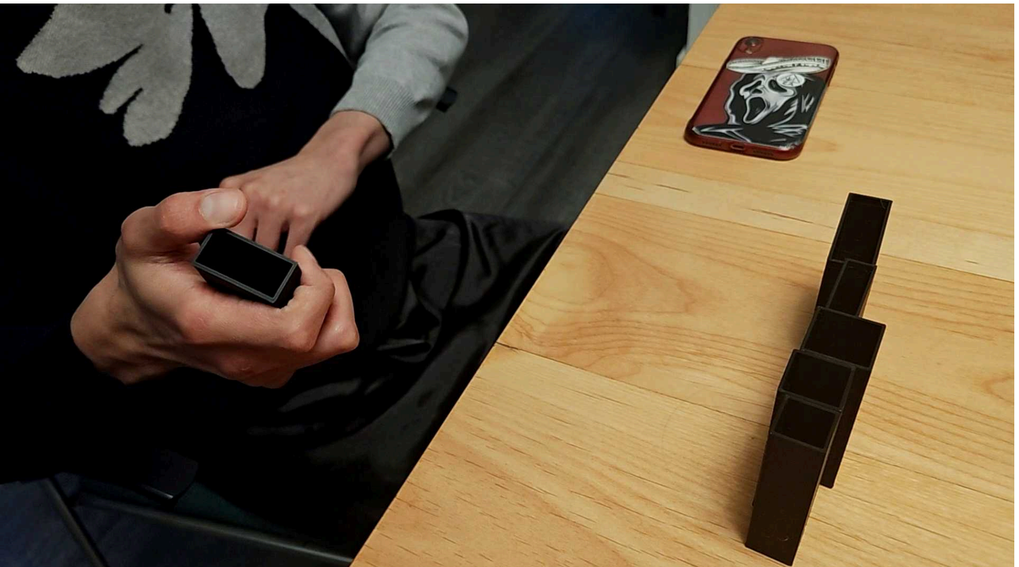
Conducted ergonomic studies of multiple enclosure sizes, determining button placement and location of hinge. Limitations of mockup hinges and buttons made interviews more difficult.

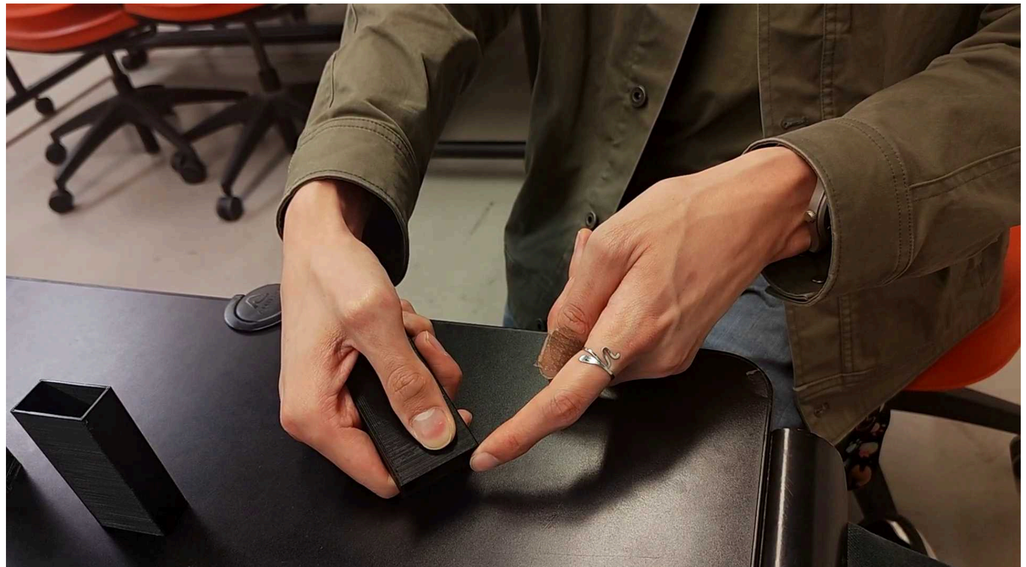
Image of available steel stocks at Bay Metals in San Francisco





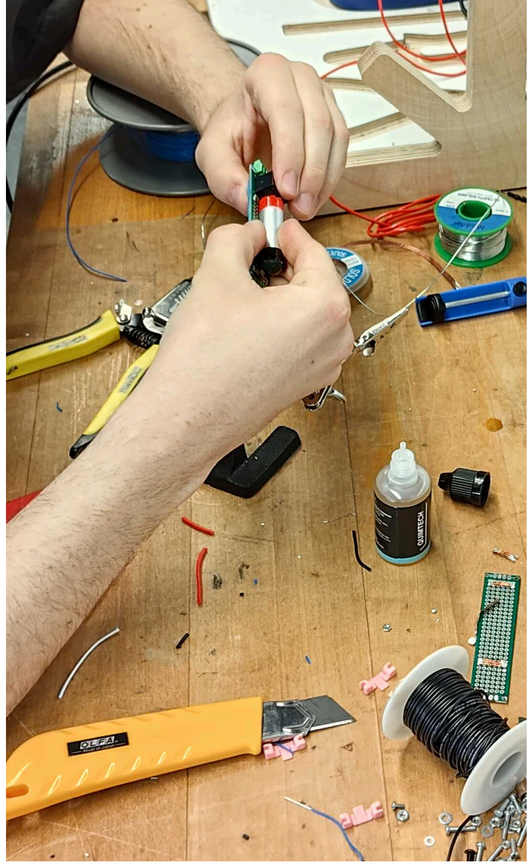
Frames from interviews for erganomics.



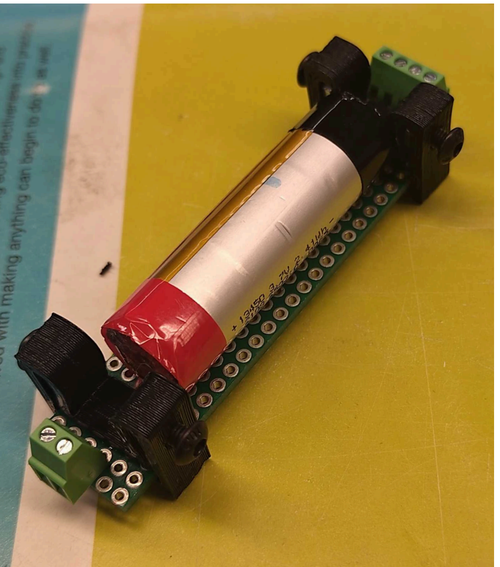


Prototyping ,development & fabrication

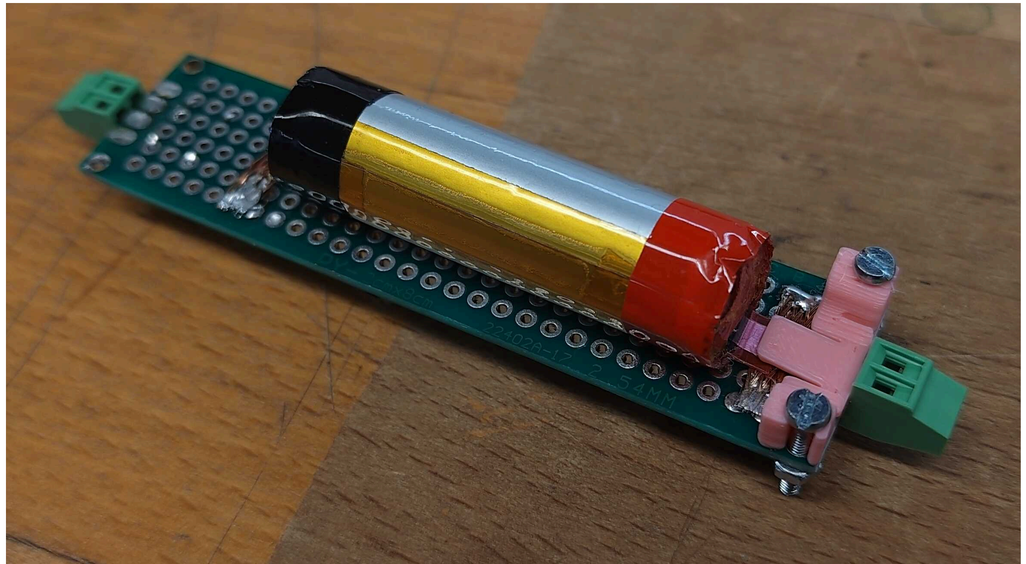
After the form factor, placement of buttons and hinges was confirmed. Prototyping and iteration began.



Assembly of close to final prototype (left). Stock modification for lid fabrication (right).



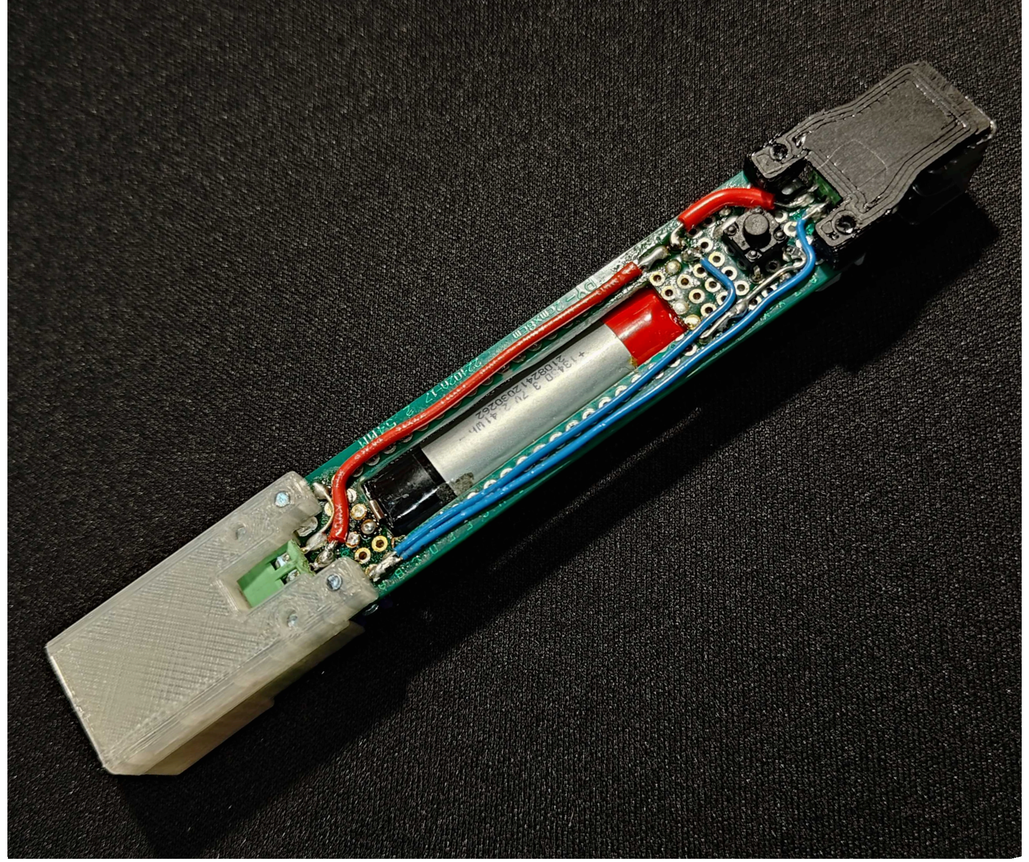
Modification of protoboard for mounting (left) Mounting markups for Delrin heating block (top) Battery assembly prototype (bottom left).

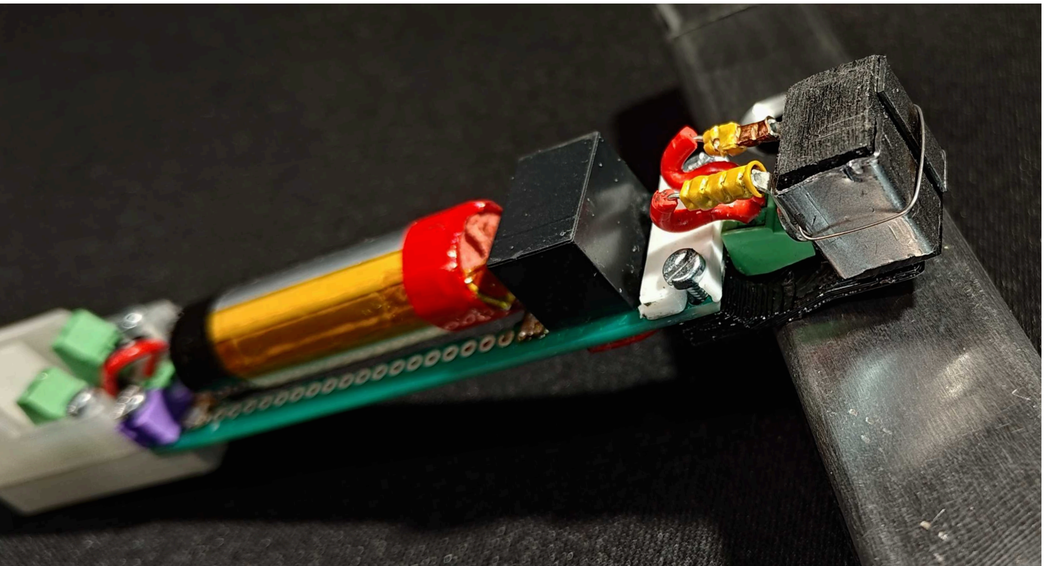


Development of final board using clips for circuit connection allowing ease of battery replacement.

Final poduct

Using a relay to switch the high current heater load, and 3d printed clips and mounts the circuit board was complete. The casign used mild steel square stock for its lid and primary housing.





Side A of internal board (Top) complete assembly (Right) Side B of internal Board(Left).